

Practice Question: The Allais Paradox

Axioms, Geometry, and Experimental Evidence

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Experimental Setup

A Laboratory Experiment

A decision-maker faces two pairwise choice problems over lotteries with real monetary payoffs. The outcome space is $X = \{\$50, \$15, \$0\}$.

Decision Problem 1: Choose between A and B .

- A : \$15 with certainty.
- B : \$50 with prob. 0.20, \$15 with prob. 0.75, \$0 with prob. 0.05.

Decision Problem 2: Choose between A^* and B^* .

- A^* : \$15 with prob. 0.25, \$0 with prob. 0.75.
- B^* : \$50 with prob. 0.20, \$0 with prob. 0.80.

Empirical Result: Suppose 68% of subjects choose A in Problem 1 *and* B^* in Problem 2. This (A, B^*) pattern is the modal choice and is inconsistent with Expected Utility Theory.

Part A: The Axiomatic Violation

- **Question A.1:** Assume a decision-maker follows Expected Utility Theory (EUT) with Bernoulli utility $u(\cdot)$. Show mathematically that a decision-maker who strictly prefers A to B must also strictly prefer A^* to B^* .
- **Question A.2:** Identify the axiom of EUT that is tested and violated by the (A, B^*) pattern. State this axiom formally for any three lotteries L, L', L'' and mixing probability $\alpha \in (0, 1)$, and show how the four lotteries in this experiment fit its structure.

Part B: Graphical Illustration (Marschak–Machina Triangle)

- **Context:** Represent this experiment in the Marschak–Machina probability triangle for the three fixed outcomes $x_H = \$50$, $x_M = \$15$, $x_L = \$0$.
- **Question B.1:** State which probabilities are measured on the horizontal axis and on the vertical axis of the triangle. Compute the coordinates (p_L, p_H) of all four lotteries (A, B, A^*, B^*) , and compute the slopes of the line segments $\overline{AB}$ and $\overline{A^*B^*}$.
- **Question B.2:** Using these slopes and the shape of EUT indifference curves in the triangle, explain why the modal (A, B^*) pattern cannot be generated by any expected-utility maximizer. Illustrate your answer with a diagram.

Part C: Behavioral Implications

- **Context:** The systematic violation of EUT in this design suggests that individuals do not evaluate lotteries linearly in probabilities.
- **Question C.1:** What psychological effect explains this pattern, and how does it change our view of what drives risk attitudes?
- **Question C.2:** Non-expected utility theories capture this with a probability weighting function $w(p)$. State the mathematical properties and boundary conditions w must satisfy on $[0, 1]$. Then, using $V(L) = \sum_i w(p_i) v(x_i)$ with $v(0) = 0$, show which condition on w is required to generate the (A, B^*) pattern.

Answer Key – Part A.1

Write $u_H = u(\$50)$, $u_M = u(\$15)$, $u_L = u(\$0)$.

$$\begin{aligned} A \succ B &\iff u_M > 0.20 u_H + 0.75 u_M + 0.05 u_L \\ &\iff 0.25 u_M > 0.20 u_H + 0.05 u_L && \text{(subtract } 0.75 u_M) \\ &\iff 0.25 u_M + 0.75 u_L > 0.20 u_H + 0.80 u_L && \text{(add } 0.75 u_L) \\ &\iff EU(A^*) > EU(B^*) \iff A^* \succ B^*. \end{aligned}$$

- Every step is an equivalence, so EUT permits only (A, A^*) or (B, B^*) (or indifference in both problems).
- The modal (A, B^*) pattern therefore contradicts EUT for every utility function u , regardless of risk attitude.

Answer Key – Part A.2

The Independence Axiom. For all lotteries L, L', L'' and all $\alpha \in (0, 1)$:

$$L \succsim L' \iff \alpha L + (1 - \alpha)L'' \succsim \alpha L' + (1 - \alpha)L''.$$

Structure of the experiment (a *common-consequence* design). Let $\alpha = 0.25$ and

$$L = \delta_{\$15}, \quad L' = Q \equiv (\$50, 0.8; \$0, 0.2).$$

- Problem 1 uses common consequence $L'' = \delta_{\$15}$: $A = 0.25 \delta_{\$15} + 0.75 \delta_{\$15}$, $B = 0.25 Q + 0.75 \delta_{\$15}$.
- Problem 2 uses common consequence $L'' = \delta_{\$0}$: $A^* = 0.25 \delta_{\$15} + 0.75 \delta_{\$0}$, $B^* = 0.25 Q + 0.75 \delta_{\$0}$.
- By independence, $A \succ B \iff \delta_{\$15} \succ Q \iff A^* \succ B^*$. Changing the common 75% component from \$15 to \$0 should not reverse the choice, yet it does for most subjects.

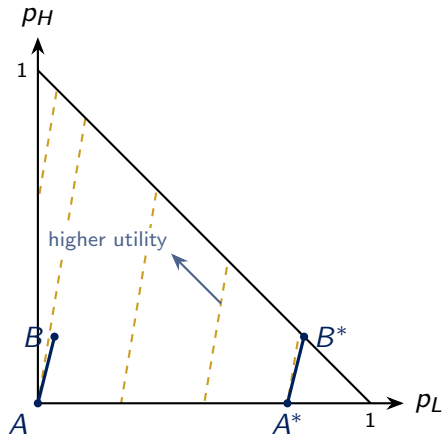
Answer Key – Part B.1 (Coordinates and Slopes)

- **Axes:** horizontal axis = p_L , the probability of the lowest outcome (\$0); vertical axis = p_H , the probability of the highest outcome (\$50). The middle outcome gets $p_M = 1 - p_L - p_H$.
- The origin is \$15 for sure; the corners (1, 0) and (0, 1) are \$0 and \$50 for sure. The hypotenuse contains lotteries with $p_M = 0$.

Lottery	p_L	p_H	p_M
A	0	0	1
B	0.05	0.20	0.75
A*	0.75	0	0.25
B*	0.80	0.20	0

- **Slopes:** $\overline{AB}: \frac{0.20 - 0}{0.05 - 0} = 4$; $\overline{A^*B^*}: \frac{0.20 - 0}{0.80 - 0.75} = 4$. The two choice segments are parallel.

Answer Key – Part B.2 (Geometric Illustration)



Dashed: EUT indifference lines with slope $6 > 4$.

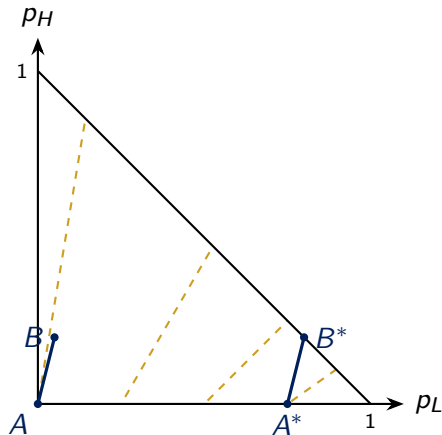
- Under EUT, $U = p_L u_L + p_H u_H + (1 - p_L - p_H) u_M$ is linear in (p_L, p_H) , so indifference curves are straight, parallel, upward-sloping lines with common slope

$$k = \frac{u_M - u_L}{u_H - u_M} > 0.$$

Utility rises toward the northwest.

- $U(B) - U(A) = U(B^*) - U(A^*) = 0.20(u_H - u_M) - 0.05(u_M - u_L)$, so $A \succ B \iff k > 4 \iff A^* \succ B^*$.
- Both choice segments have slope 4, so one common slope k ranks both pairs the same way: steep lines ($k > 4$) give (A, A^*) ; flat lines ($k < 4$) give (B, B^*) .
- (A, B^*) would need $k > 4$ near A and $k < 4$ near A^* : impossible with parallel lines.

Answer Key – Part B.2 (Remark: Fanning Out)



Dashed: non-EUT indifference curves that fan out.

- The (A, B^*) pattern *can* be generated if indifference curves are not parallel.
- Near A (the northwest, better lotteries) the curves are steeper than 4, so $A \succ B$.
- Near A^* (the southeast, worse lotteries) they are flatter than 4, so $B^* \succ A^*$.
- This is Machina's **fanning-out hypothesis**: people become more risk averse as lotteries improve. It is the geometric signature of the common-consequence effect and is consistent with probability weighting (Part C).

The certainty effect (Kahneman & Tversky, 1979).

- People overweight outcomes that are certain relative to outcomes that are merely probable.
- Problem 1: moving from A to B introduces a 5% chance of \$0 into a sure thing. Losing certainty feels large, so most choose A .
- Problem 2: moving from A^* to B^* raises an already large chance of \$0 from 75% to 80%. This feels negligible, so most chase the \$50 in B^* .
- Under EUT the two changes are identical (the same 0.05 shift in p_L and 0.20 shift in p_H), so they must be weighed the same way.

Implication for risk attitudes: risk aversion is not determined only by the curvature of the utility function over money. It also depends on how people *perceive and weight probabilities*, which is nonlinear and especially sensitive near 0 and 1.

Answer Key – Part C.2

Required properties of $w : [0, 1] \rightarrow [0, 1]$:

- Strictly increasing (and typically continuous), with boundary conditions $w(0) = 0$ and $w(1) = 1$.
- EUT is the special case $w(p) = p$. Empirically, w is *inverse-S shaped*: small probabilities are overweighted and moderate-to-large ones underweighted.

Generating (A, B^*) with $V(L) = \sum_i w(p_i) v(x_i)$, $v(0) = 0$:

$$\begin{aligned} A \succ B &\iff v(15) > w(0.20) v(50) + w(0.75) v(15) \iff [1 - w(0.75)] v(15) > w(0.20) v(50), \\ B^* \succ A^* &\iff w(0.20) v(50) > w(0.25) v(15). \end{aligned}$$

Combining: $w(0.25) v(15) < w(0.20) v(50) < [1 - w(0.75)] v(15)$, which requires

$$w(0.25) + w(0.75) < 1 \quad (\text{subcertainty}).$$

With $w(p) = p$ this sum equals 1, which is why EUT cannot produce the pattern.